

# Analyzing Trends in Technology

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# OUTLINE

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- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

# EXECUTIVE SUMMARY

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- Current Technology Usage
- Future Technology Trends
  - Programming Language
  - Database
- Demographics

# INTRODUCTION

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## Overview

- This presentation uses data analytics to explore trends in in-demand IT skills.
- Focus is on current and future needs in:
  - Programming Languages
  - Database Technologies
  - IDEs and Web Frameworks

## Research Objectives

- Identify the most in-demand programming languages.
- Determine which database skills are highly sought after.
- Highlight popular IDEs and web frameworks used in the industry.

## Target Audience

- IT professionals
- HR managers and recruiters
- Anyone interested in understanding evolving IT skill demands

# METHODOLOGY

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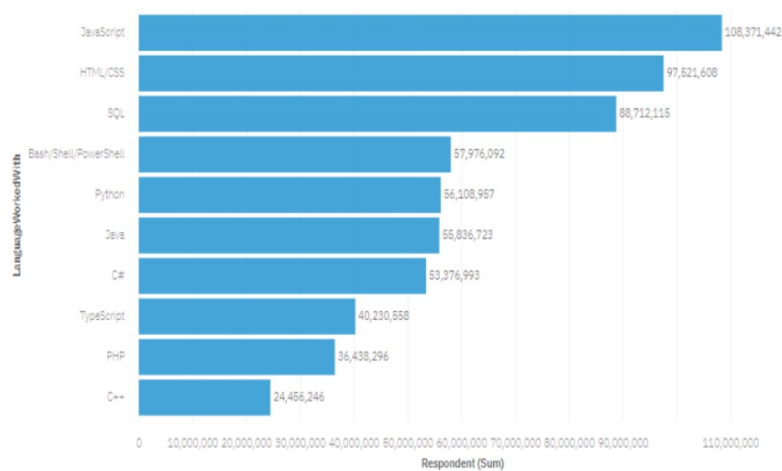


- Data collection
- Data Exploration
- Data Wrangling
- Data Visualization
- Presentation

# PROGRAMMING LANGUAGE TRENDS

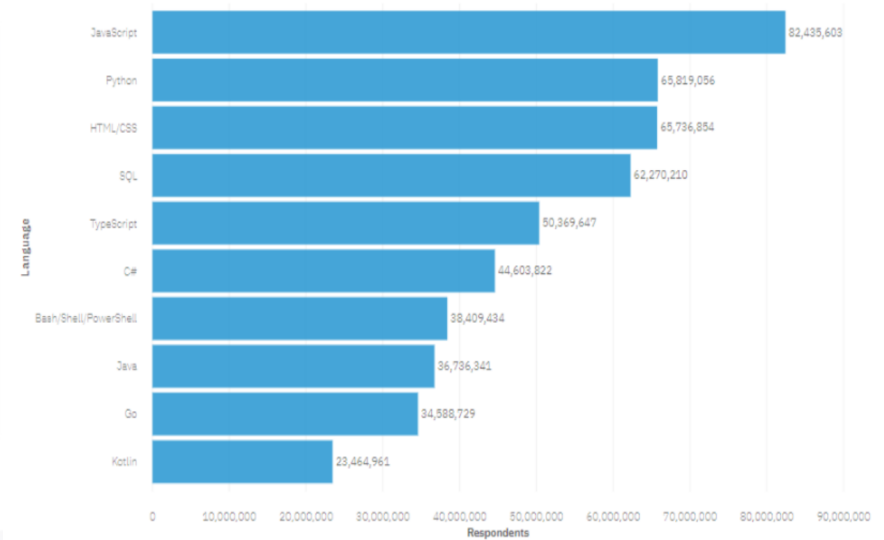
Current Year

Top 10 Languages



Next Year

Top 10 Language Desire Next Year



# PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

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## Findings

### Current Year - Top 10 Languages (by usage)

- JavaScript, HTML/CSS, and SQL are the top 3 most widely used languages.
- Bash/Shell/PowerShell and Python also have strong usage numbers, followed closely by Java and C#.
- TypeScript, PHP, and C++ round out the top 10.

### Next Year - Top 10 Languages by Developer Interest

- JavaScript remains the most desired skill, showing consistency.
- Python and HTML/CSS are highly desired for future learning, indicating growing interest.
- TypeScript and Go show upward momentum in developer interest.
- Kotlin enters the top 10 future-interest list, even though it's not in the current top 10 usage list.

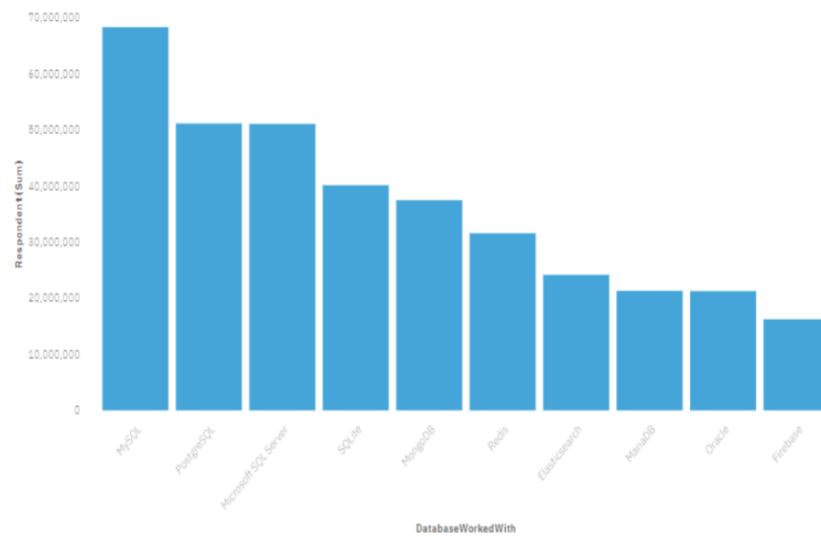
## Implications

- JavaScript's dominance in both charts suggests it will remain a critical language to learn and use, both now and in the near future.
- Python's high future demand suggests it's increasingly seen as a versatile language (data science, web, AI), even though its current usage is slightly lower.
- TypeScript is growing in relevance — likely due to its strong adoption in modern web development.
- Languages like Go and Kotlin, though less used now, show rising interest, pointing to emerging trends in backend systems and Android development.
- IT professionals and HR teams should prioritize hiring and training in JavaScript, Python, and SQL, while also preparing for the rising demand in modern tools like TypeScript and Go.

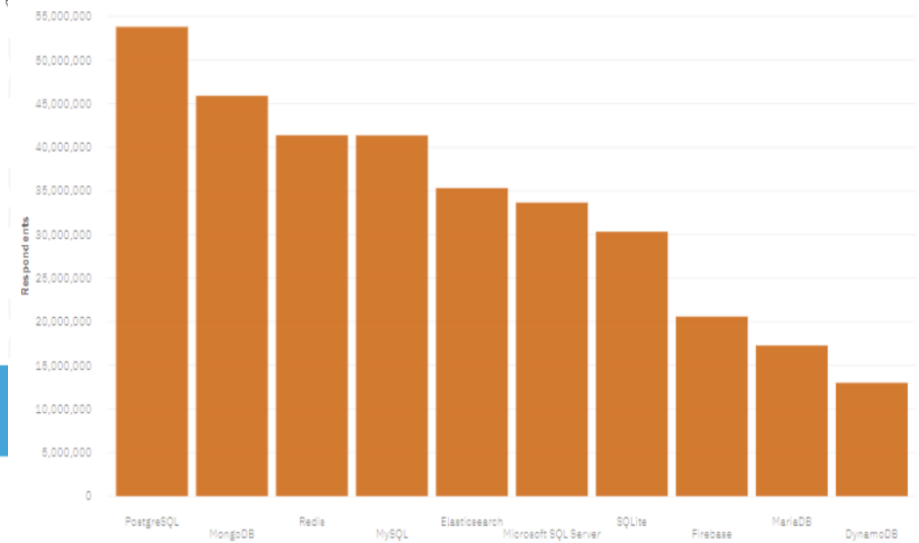
# DATABASE TRENDS

## Current Year

Top 10 Databases



## Next Year





# DATABASE TRENDS - FINDINGS & IMPLICATIONS

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## Findings

- MySQL remains the most widely used database at present.
- PostgreSQL and MongoDB are becoming increasingly popular choices.
- Elasticsearch is also seeing a steady rise in adoption.

## Implications

- Open-source databases continue to be widely favored across the industry.
- NoSQL databases are predominantly utilized for storing non-relational data.
- Technologies like Redis and Elasticsearch show strong potential to shape the future of database solutions.

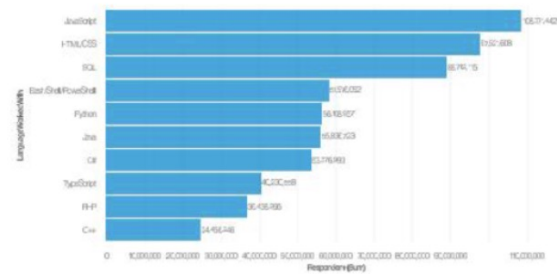
# DASHBOARD

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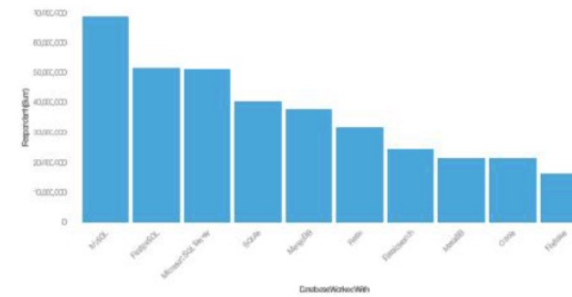


# DASHBOARD TAB 1

Top 10 Languages



Top 10 Databases



Platform worked with currently

Respondents  
100% 100%



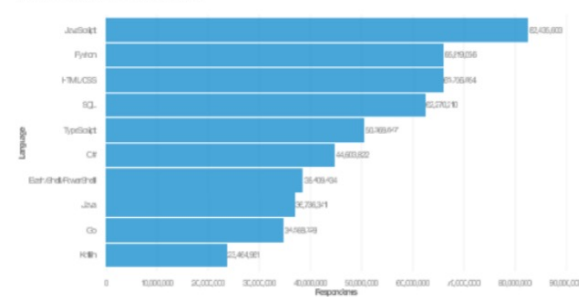
Top 10 WebFrame worked with Currently

Respondent (Sum)  
100% 100%

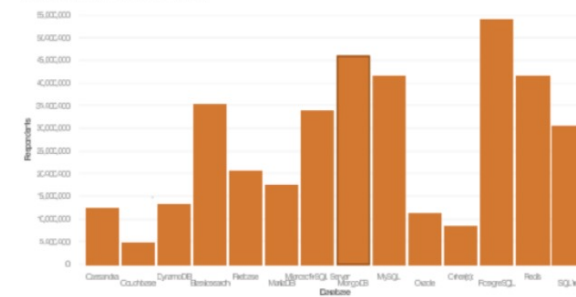


# DASHBOARD TAB 2

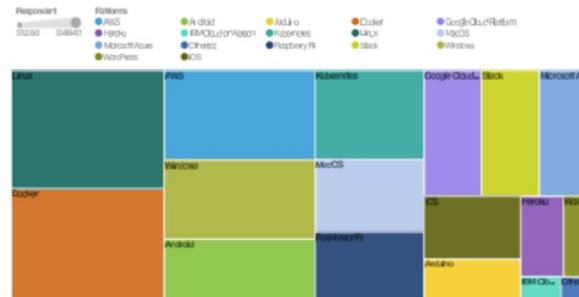
Top 10 Language Desire Next Year



Top 10 Database Desire Next Year



Platform Desire Next Year.



Top 10 Web Frame Desire Next Year



# DASHBOARD TAB 3

Percentage of Respondents by gender

Gender  
● Woman ● Man

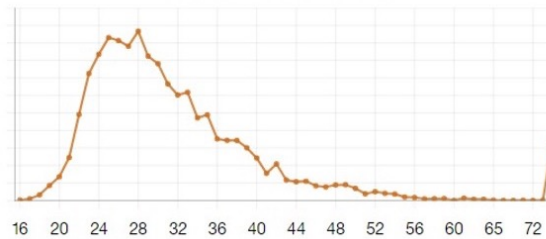


Respondents by Country

Respondent (Sum)

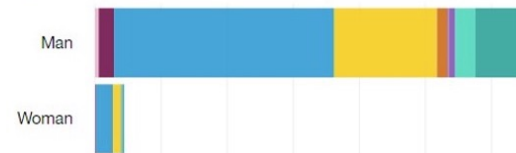


Respondents by Age



Respondents by Formal Education Level

Education level  
● (no value)  
● Associate degree



# DISCUSSION

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# OVERALL FINDINGS & IMPLICATIONS

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## Findings

- Technology in the field is evolving rapidly.
- The majority of individuals in the IT industry hold a Bachelor's degree.
- Web development languages remain highly popular and in demand.
- There is a noticeable concentration of professionals in countries like the USA and India.
- There exists a significant age and gender disparity within the tech workforce.
- The sector is predominantly composed of young professionals under the age of 40.
- Tools like AWS and Docker are seeing increased adoption.
- Many respondents plan to learn PostgreSQL and React JS in the upcoming year.

## Implications

- Data professionals should consider gaining skills in NoSQL alongside traditional SQL databases.
- Organizations need to stay updated with and adopt emerging technologies.
- Web development continues to be a valuable and profitable skill set.
- Programs aimed at reducing the age and gender gaps in tech should be introduced.
- There is a need to boost recruitment in roles focused on emerging technologies.
- Expanding access to tech education and training in underdeveloped countries is crucial.
- Businesses should leverage tools like Docker and cloud services (e.g., AWS) for efficient application deployment.

# CONCLUSION

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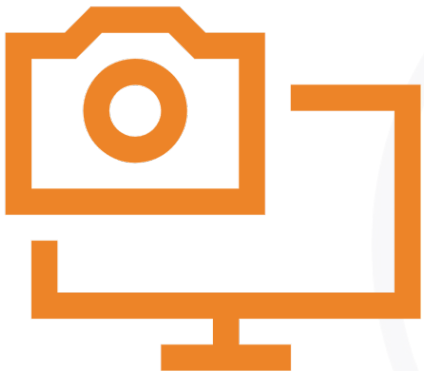
Staying current with trends in the tech industry is essential due to its constant evolution.

Keeping up with emerging tools and technologies ensures continued relevance and competitiveness in the field.



# APPENDIX

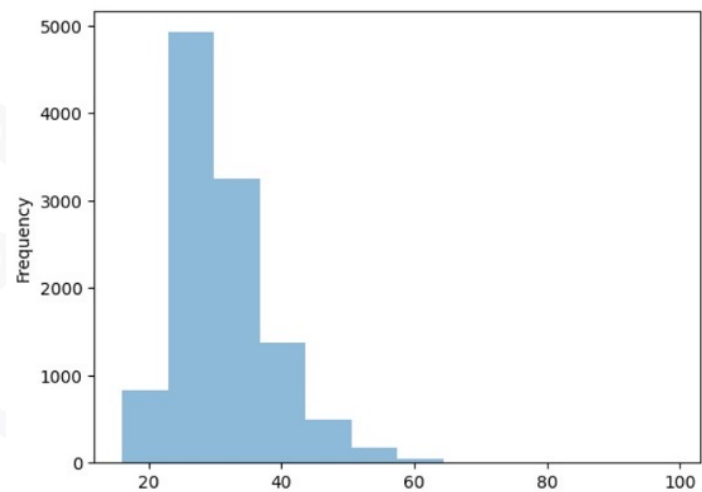
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Plot a histogram of the column `Age`.

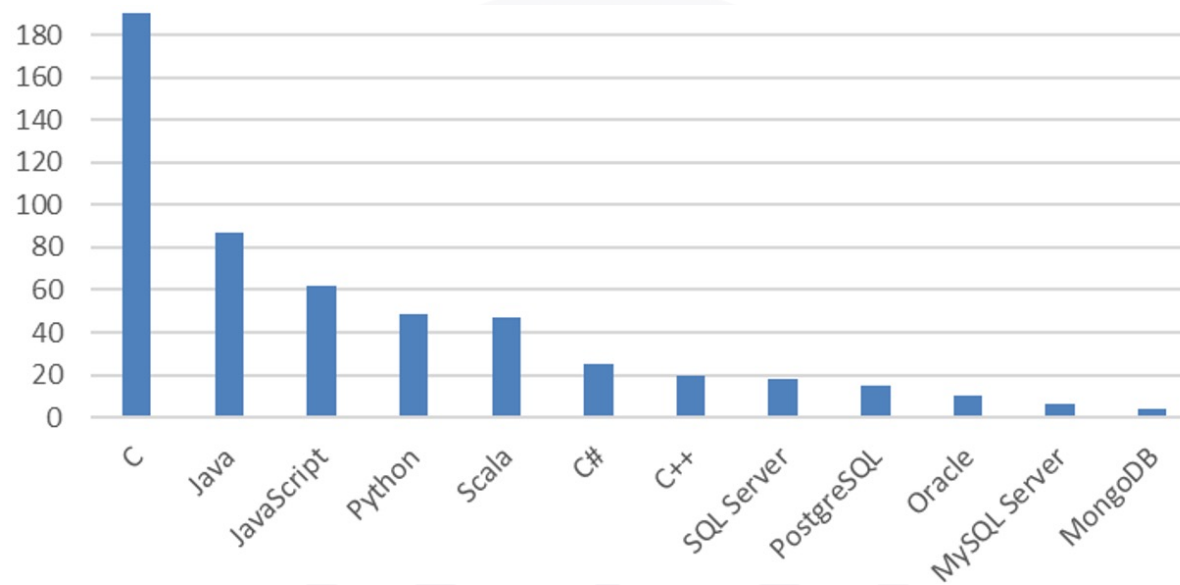
```
df['Age'].plot.hist(bins=12, alpha=0.5) #
```

<AxesSubplot: ylabel='Frequency'>



# JOB POSTINGS

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# POPULAR LANGUAGES

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